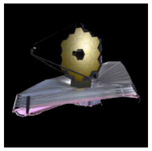

7677 - Pushing the Faintest Limits: Extremely Low-Luminosity and and Pop III-like Star-Forming Complexes in the Early Universe

Cycle: 4, Proposal Category: GO

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OBSERVATIONS

Folder	***Observation***	***Label***	***Observing Template***	***Science Target***
IFU - LAP2 - BOTTOM				
	1	LAP2-bottom	NIRSpec IFU Spectroscopy	(1) LAP2-bottom
LAP2-top				
	2	LAP2-top	NIRSpec IFU Spectroscopy	(2) LAP2-top

ABSTRACT

We propose JWST NIRSpec/IFU spectroscopy of two extraordinarily faint and strongly lensed Ly-alpha emitters at z=5.663 and z=4.194, each straddling their respective critical lines. The primary target at z=5.663 (dubbed LAP2) is an exceptional ionizing source with a remarkably large Ly-alpha equivalent width (>320 A rest-frame) and an extremely faint ultraviolet magnitude (Muv > -11.6, delensed > 35). LAP2 remains undetected in all NIRCam bands, including stacked SW and LW images, suggesting an ultra-low stellar mass of < 10^4 Msun. Its identification as an isolated, low stellar mass star-forming complex within the first Gyr offers an unparalleled opportunity to investigate star formation in near-pristine conditions. The secondary target at z=4.194, also straddling its critical line, has Muv ~ -12.3 and is detected in NIRCam with a de-lensed size of <10 pc, potentially resembling a proto-globular cluster hosting massive O-type stars. This study will allow us to measure the metallicity of both sources and assess the presence of massive stars in such elusive systems by evaluating their ionizing photon production efficiency. These observations will expand (at least double) the sample of ultra-faint sources with these measurements which only JWST can perform, pushing the frontier of understanding toward Population III-like star formation conditions. The fortunate angular proximity of the two targets allows for simultaneous observation within the same IFU field of views.

OBSERVING DESCRIPTION

The program aims to observe two Ly-alpha emitters straddling their respective critical lines in a strongly lensed field at z=4.194 and z=5.663 (the latter referred to as LAP2).

Based on the Ly-alpha fluxes, we will detect Balmer lines (and potentially other metal lines, if present). To optimize the search for these emissions, we will employ NIRSpec/IFU in the clear/prism mode, which covers the full spectral range from Ly-alpha to H-alpha. The low spectral resolution R~100 is appropriate for the scientific objectives of this proposal.

Observational strategy and depth:

The angular proximity of the two Ly-alpha arclets allows them to be captured within the same NIRSpec IFU field of view. However, due to the extension of the arclets, two slightly overlapping pointings are required. More than 90% of the spaxels will benefit from magnification factors exceeding ×50-100.

The sensitivity is calibrated to probe low metallicity conditions ($Z < 0.01$ Zsun) for both targets, 6.8h integration per pointing. We propose using PRISM/CLEAR with NRSIRS2, configured for 21 groups and 2 integrations. The dither pattern will be cycling with a small 8-point configuration.

The total requested NIRSpec IFU time is 17.4 hours, including overheads.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(1)	LAP2-bottom	RA: 00 14 21.8303 (3.5909596d) Dec: -30 23 56.15 (-30.39893d) Equinox: J2000		
Comments: BOTTOM part of the pair: This IFU pointing include stwo super-magnified Ly-alpha arclets at z=5.663 and z=4.194 straddling the critical lines. Gravitational lensing allow us to probe intrinsic magnitude fainter than 34 (Muv -13) up to 37 (Muv -10). *Category=Galaxy* *Description=[High-redshift galaxies, Lyman-alpha galaxies]*				
(2)	LAP2-top	RA: 00 14 21.9504 (3.5914600d) Dec: -30 23 55.38 (-30.39872d) Equinox: J2000		
Comments: TOP part of the pair: This IFU pointing include stwo super-magnified Ly-alpha arclets at z=5.663 and z=4.194 straddling the critical lines. Gravitational lensing allow us to probe intrinsic magnitude fainter than 34 (Muv -13) up to 37 (Muv -10). *Category=Galaxy* *Description=[High-redshift galaxies, Lyman-alpha galaxies]*				

Observation

Proposal 7677, Observation 1: LAP2-bottom

Tue Mar 11 21:03:07 GMT 2025

Diagnostic Status: Warning

Observing Template: NIRSpec IFU Spectroscopy

Diagnostics

(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(1)	LAP2-bottom	RA: 00 14 21.8303 (3.5909596d) Dec: -30 23 56.15 (-30.39893d) Equinox: J2000		

Comments: BOTTOM part of the pair: This IFU pointing include stwo super-magnified Ly-alpha arclets at z=5.663 and z=4.194 straddling the critical lines. Gravitational lensing allow us to probe intrinsic magnitude fainter than 34 (Muv -13) up to 37 (Muv -10).
Category=Galaxy
Description=[High-redshift galaxies, Lyman-alpha galaxies]

Template

TA Method	HFF Readout Mode
NONE	false

Dithers

#	Dither Type	Size	Starting Point	Number of Points	Points
1	CYCLING	SMALL	1	8	

Spectral Elements

#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
1	PRISM/CLEAR	NRSIRS2	21	2	false	true	NONE	8	16	24742.757	

Observation

Proposal 7677, Observation 2: LAP2-top

Tue Mar 11 21:03:07 GMT 2025

Diagnostic Status: Warning

Observing Template: NIRSpec IFU Spectroscopy

Diagnostics

(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(2)	LAP2-top	RA: 00 14 21.9504 (3.5914600d) Dec: -30 23 55.38 (-30.39872d) Equinox: J2000		

Comments: TOP part of the pair: This IFU pointing include stwo super-magnified Ly-alpha arclets at z=5.663 and z=4.194 straddling the critical lines. Gravitational lensing allow us to probe intrinsic magnitude fainter than 34 (Muv -13) up to 37 (Muv -10).
Category=Galaxy
Description=[High-redshift galaxies, Lyman-alpha galaxies]

Template

TA Method	HFF Readout Mode
NONE	false

Dithers

#	Dither Type	Size	Starting Point	Number of Points	Points
1	CYCLING	SMALL	1	8	

Spectral Elements

#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
1	PRISM/CLEAR	NRSIRS2	21	2	false	true	NONE	8	16	24742.757	